

Adarsh ML, DL, NLP
Course

①

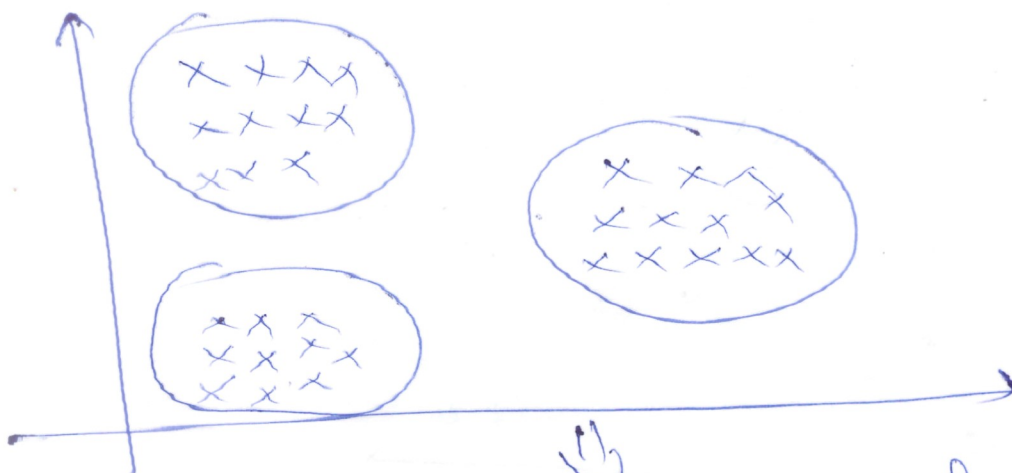
Krishna sir

Sec 4:- sec 4/
Hierarchical
clustering

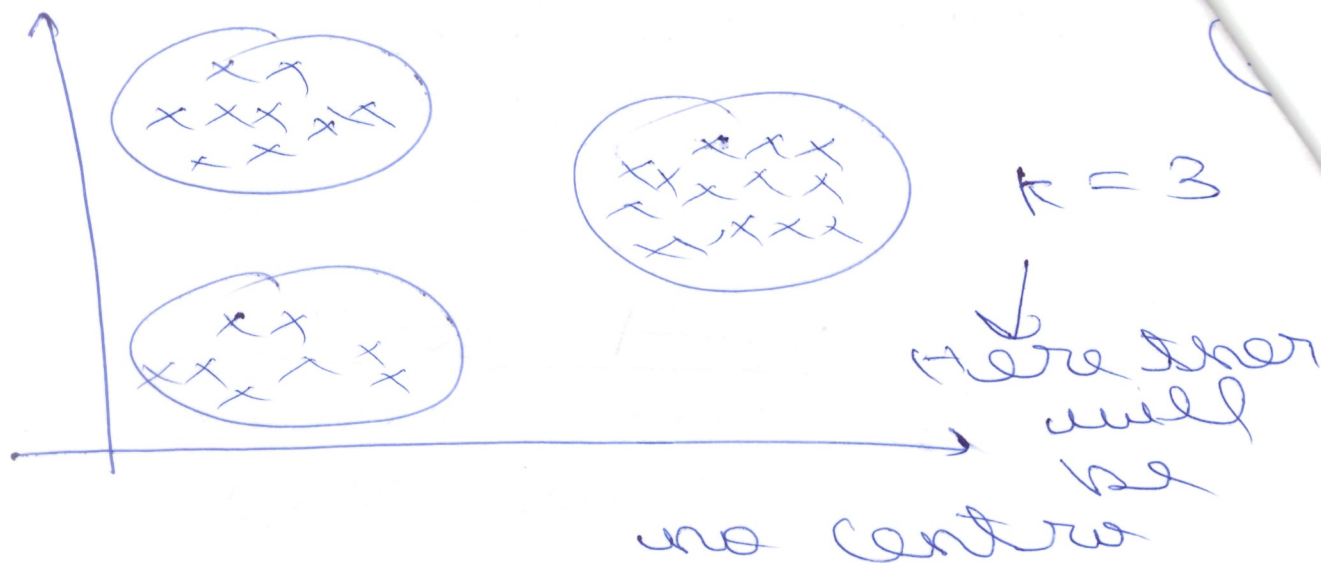
lec 1:- Hierarchical
clustering
→

lec 2:- K means vs
Hierarchical
clustering

Suppose the dataset is
given in the form as:-



↓
After applying
Hierarchical clustering

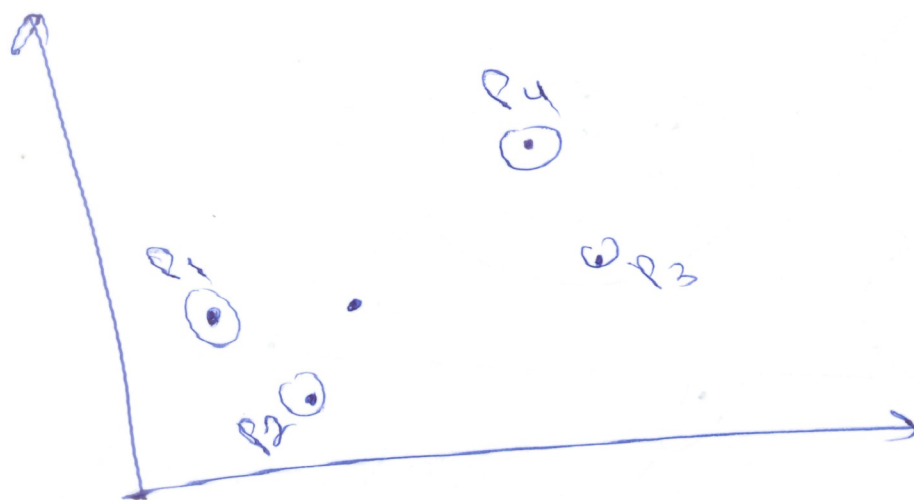


Hierarchical clustering

① Agglomerative

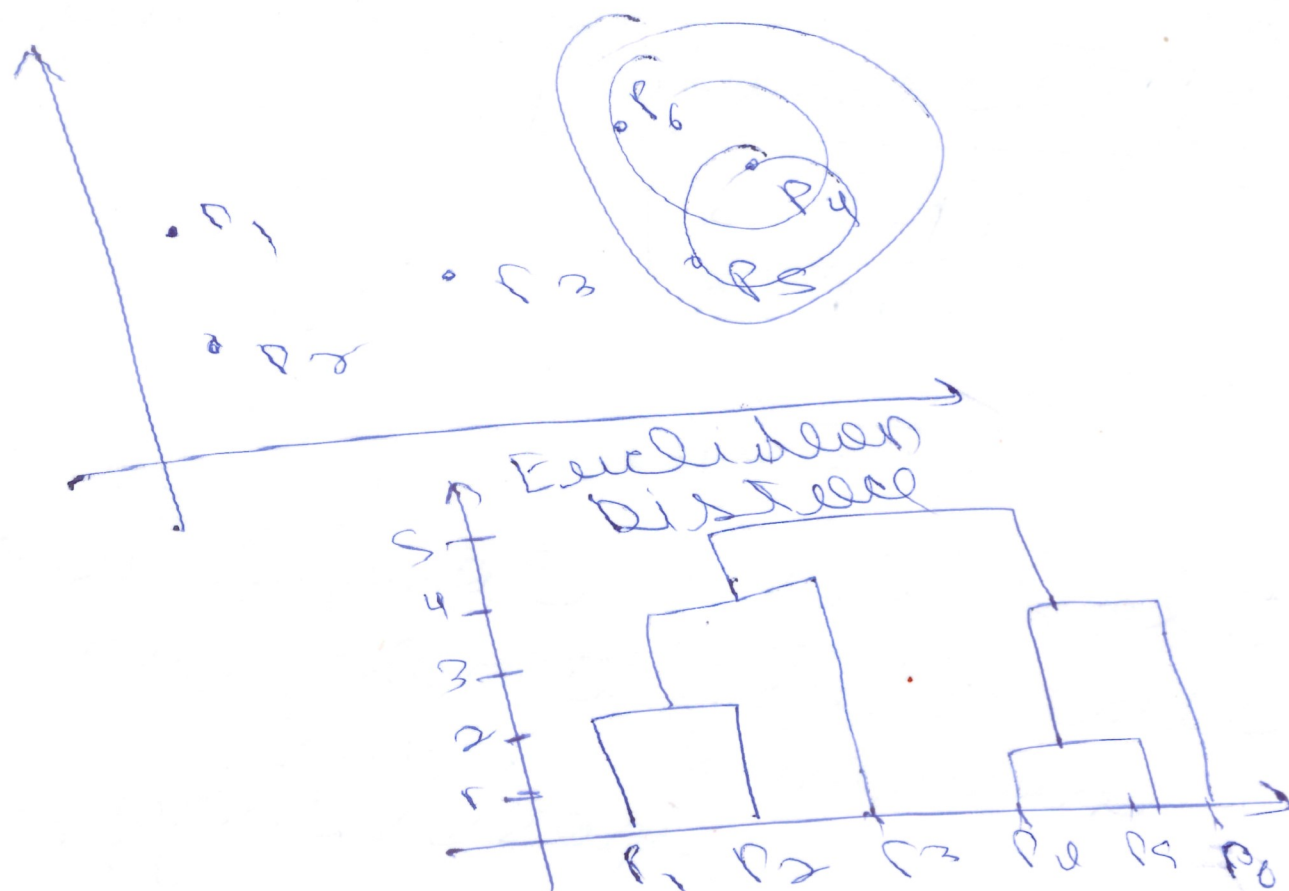
② divisive

Geometric

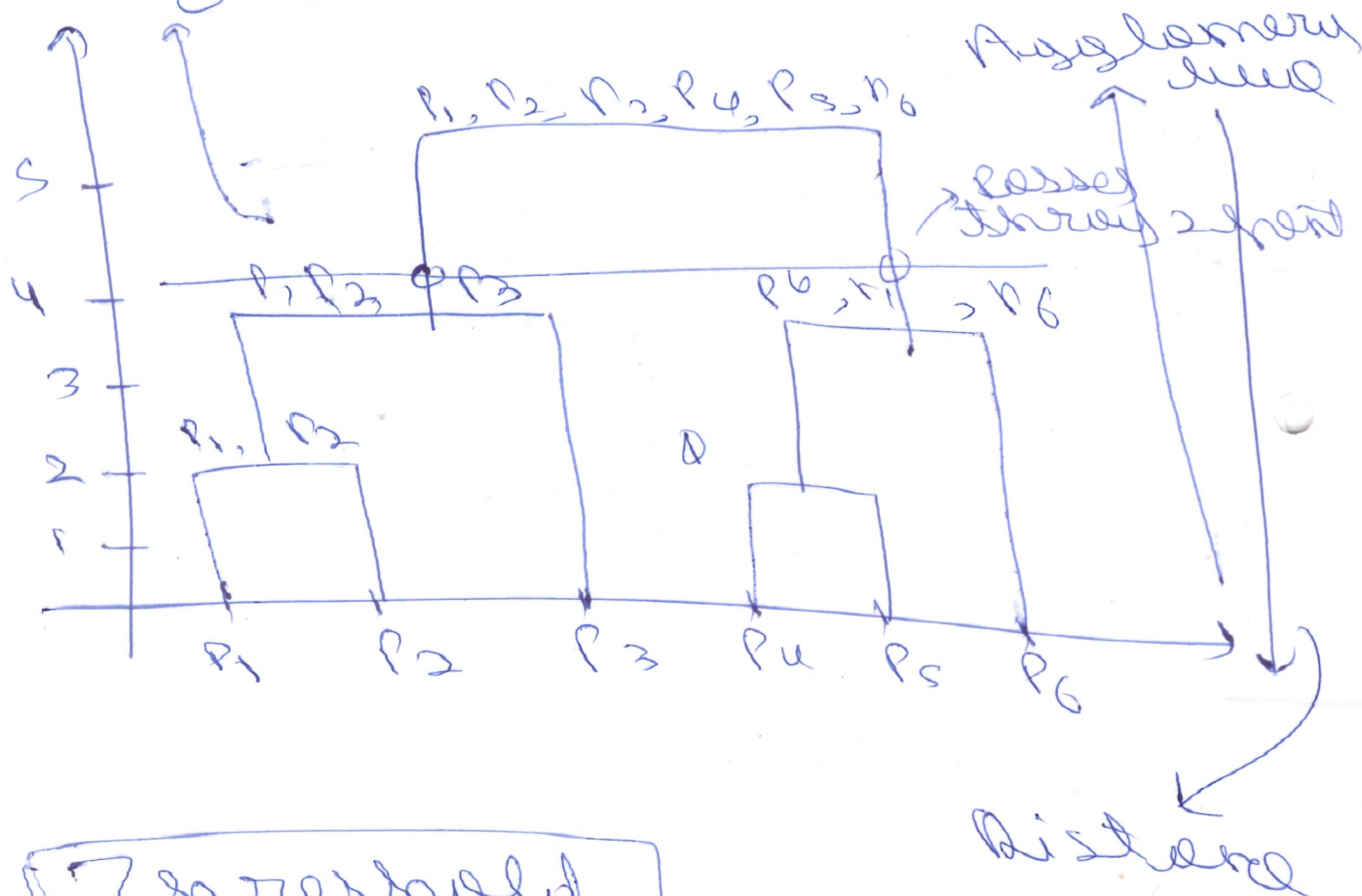


- ① For each data point, initially will consider it as a separate cluster
- ② Find the nearest point and create a new cluster
- ③ Keep on doing the same process, until we get a single cluster.

a dendrogram



Then, the no. of clusters, $k=2$, can be calculated with threshold = 4



Threshold

$k=2$

$k=4$

Select the longest with
one such that no
horizontal lines
pass through it

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K means vs Hierarchical clustering

Scalability and flexibility

(1) Dataset size →

Large → K means
Small → Hierarchical clustering

(2) K means → only for numerical data.

Hierarchical clustering → can handle data along with numerical one

(3) Centroids used in K-means

(4) In Hierarchical, we use no. of clusters.